Spatial-dependence sensitivity of the H1 analysis

Response to the reviewer

We agree that spatial dependence among pollen records can make densely sampled areas disproportionately influential. We therefore added a reproducible sensitivity analysis that combines spatial subsampling with explicit spatial control.

The models contributing to Figure 2 are fitted separately within pollen cores. Coordinates are constant within a core and therefore cannot be included as a within-core predictor. We instead modelled spatial structure in the resulting core-level human-minus-climate importance balances. Positive distance-based Moran eigenvector maps (dbMEMs) were constructed independently within continents and selected conditional on the existing continent-by-climate-zone strata. We diagnosed Moran's I before and after filtering at 250 km and 500 km.

As a design-based sensitivity, we also repeated the Figure 2 aggregation after 100 random within-stratum thinnings at minimum distances of 250 km and 500 km. We additionally omitted each continent and each climate zone in turn. The classification was specified in advance: robust conclusions require agreement of the spatially filtered and deterministic estimates plus at least 95% agreement among thinning replicates.

The signed overall balance changed only from -0.529 to -0.527 after dbMEM filtering. Climate remained the larger contribution in 100% of 250-km replicates and 100% of 500-km replicates, and in every deterministic leave-out analysis. Both signed and zero-truncated profiles therefore meet the predeclared **robust** ranking criterion.

Profile	Classification	Minimum thinning agreement	Residual dependence detected
signed	robust	100%	TRUE
zero_truncated	robust	100%	TRUE

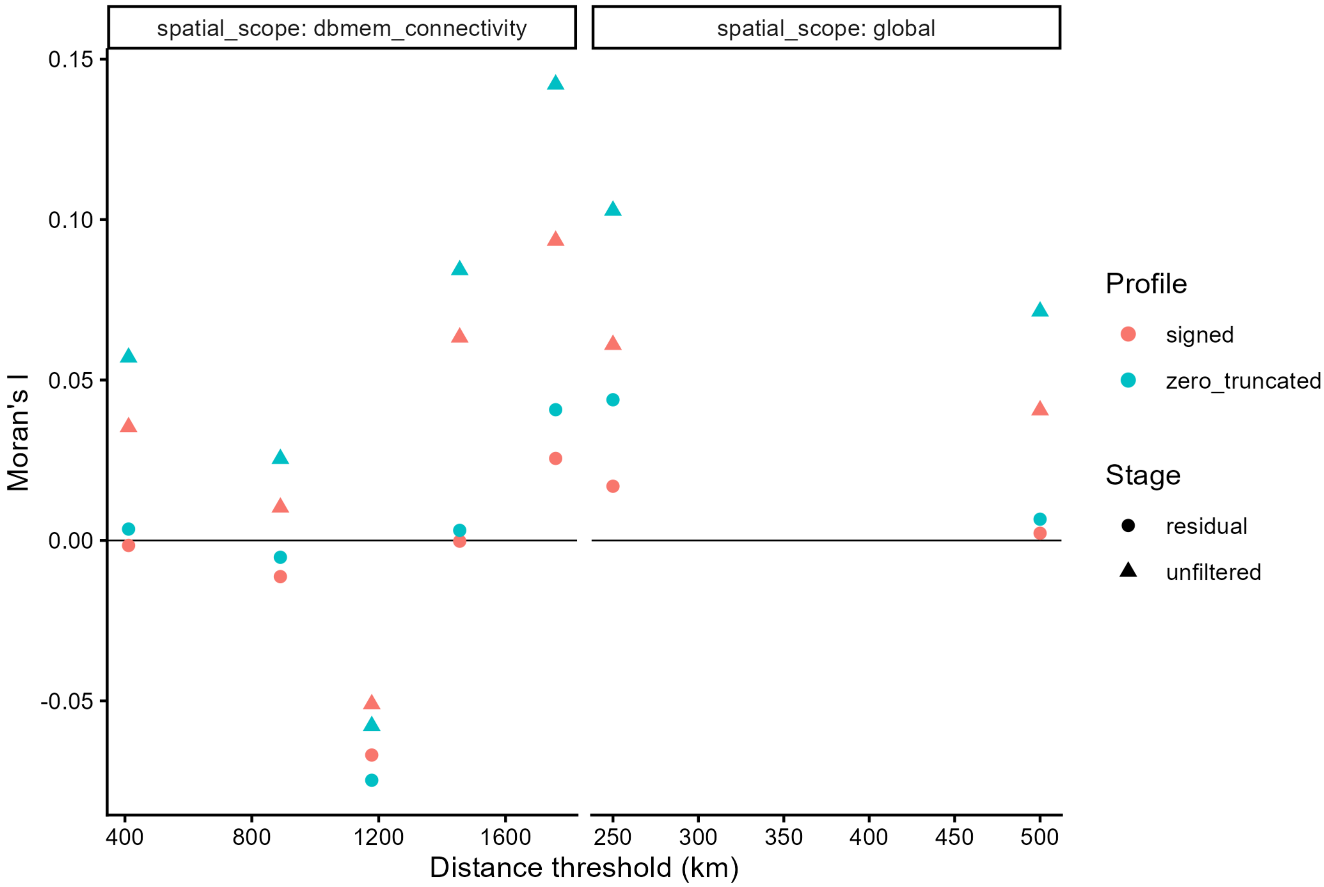

Figure 1: Spatial autocorrelation before and after dbMEM filtering.

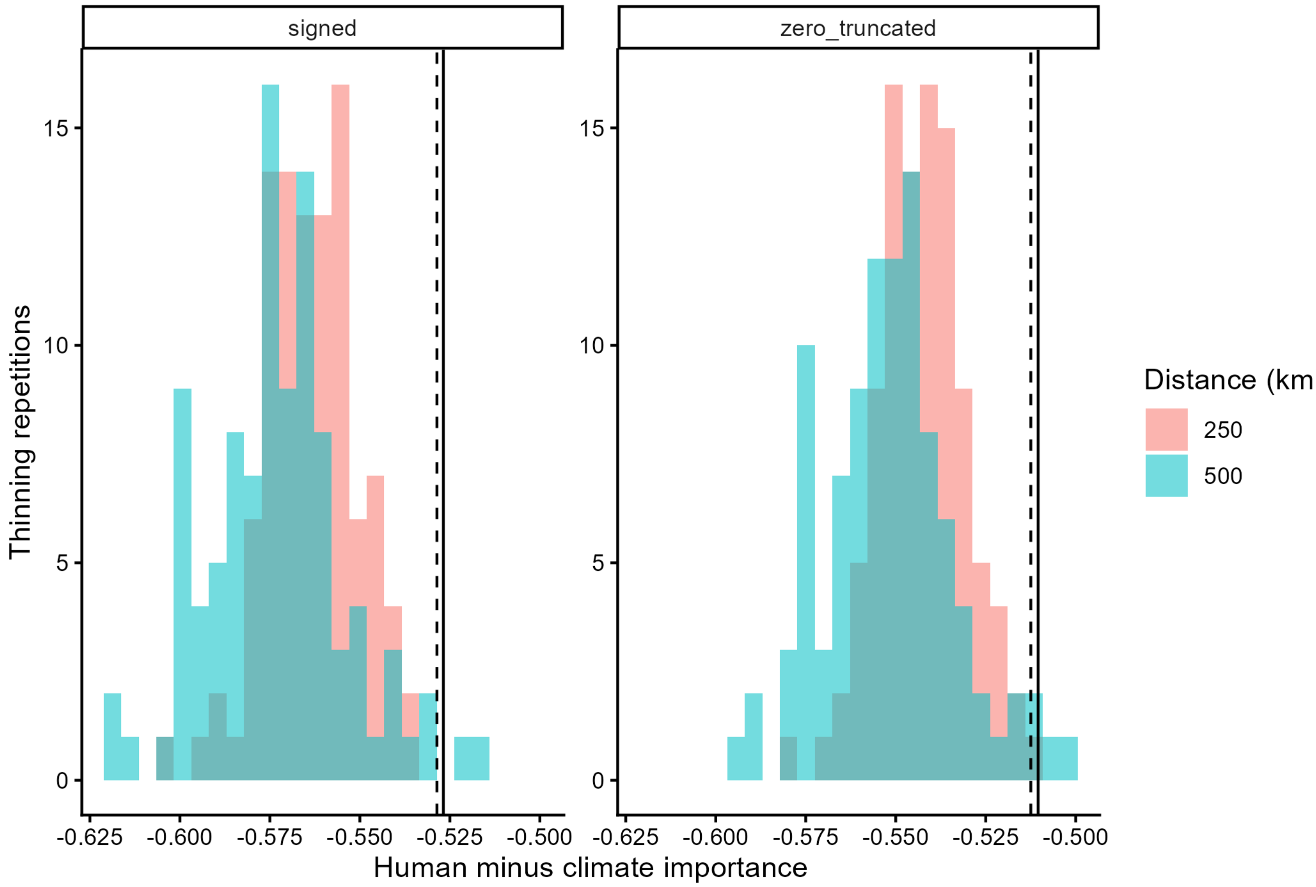

Figure 2: Human-minus-climate balance after repeated spatial thinning.

For the region-by-age analysis, sites occur together within each model, so space can be included directly. We selected dbMEMs conditional on the complete human and climate predictor sets, fitted a three-group hierarchical partition, and separately calculated pure human, pure climate, and pure spatial adjusted R^2 fractions. Hierarchical contributions and pure conditional fractions are reported separately because they answer different questions.

Spatial terms were selected in 53 of 70 region-by-age models. Detectable positive residual autocorrelation remained on at least one diagnostic axis and distance scale in 53 groups. Likewise, the Figure 2 residuals retained short-range autocorrelation at 250 km, although not at 500 km. We therefore interpret the ranking as robust to the implemented spatial controls, but do not claim that the data or residuals are spatially independent.

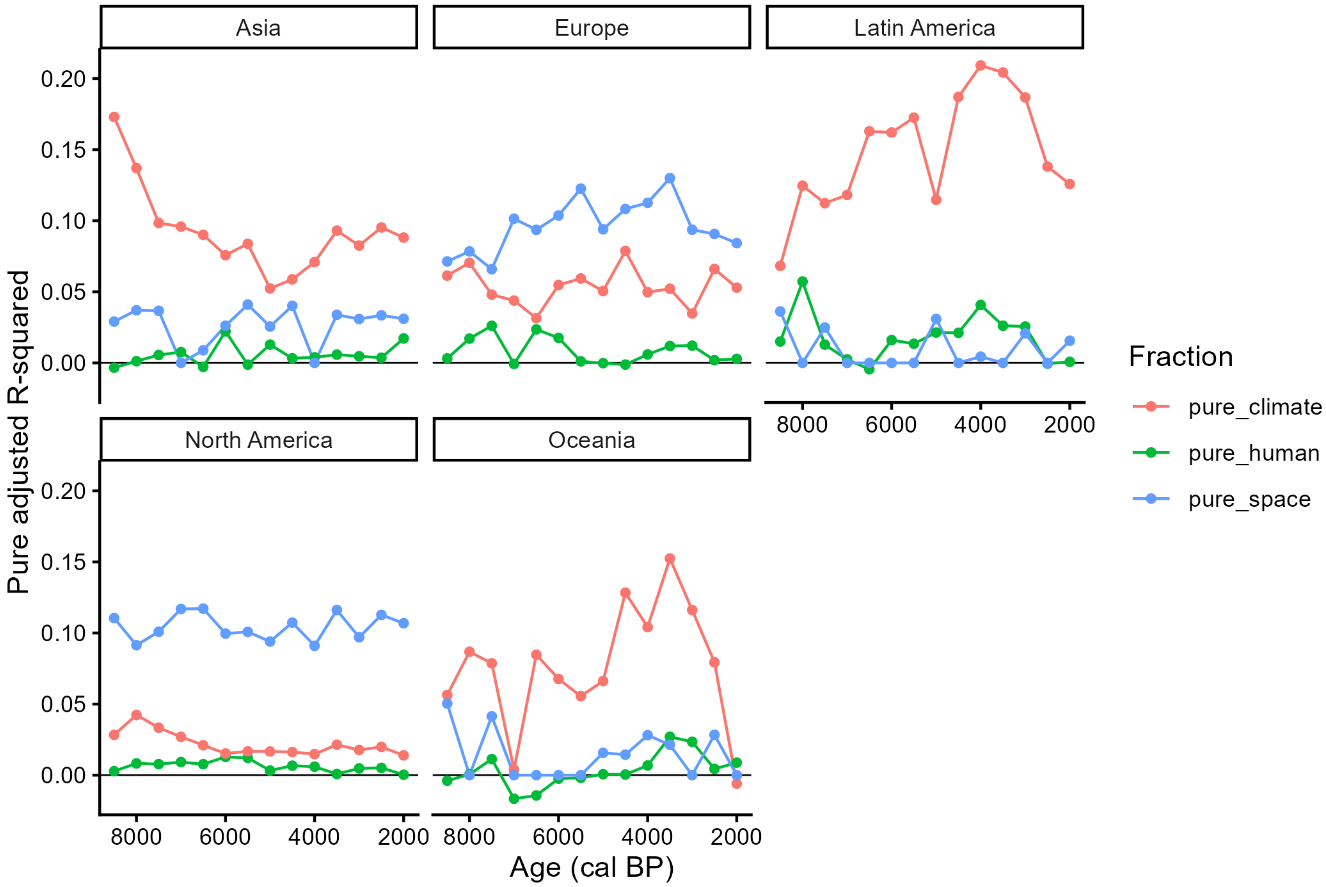

Figure 3: Pure human, climate, and spatial fractions through time.

Residual spatial dependence is reported rather than treated as automatically removed. Any region-age group with insufficient locations, rank limitations, or detectable residual autocorrelation remains explicitly flagged in the exported diagnostics.